IREF - Warming Scenarios

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Climate change is expected to have complex effects on predator-prey interactions, in part because temperature influences a broad range of organism-level processes (e.g., metabolism, growth, reproduction, and mortality) that can in turn impact how predators and their prey interact (Brown et al. 2004; Dell et al. 2011; Rall et al. 2010). Warming's potential impacts on elements of the consumptive effects (CEs) of predators on their prey (e.g., the rate at which predators encounter, subdue, and consume their prey) have been a focal point of recent research (e.g., Brose et al. 2012; Dell et al. 2014; Romero et al. 2018). However, it remains unclear how warming may impact components of predation risk effects, or "nonconsumptive" predator effects (NCEs), which have been argued to be as important or even more important to populations, communities, and food webs than their consumptive effects (Werner and Peacor 2003; Peckarsky et al. 2008; Preisser et al. 2005).

Predicting how warming will impact predation risk effects may be especially challenging, because risk effects involve a chain of complex, context-dependendent, and multi-faceted processes, each of which may be sensitive to temperature to differing degrees (Angilletta 2009; Dell et al. 2011). Broadly defined, predation risk effects involve the process by which prey respond to their predators to mitigate their predation risk, usually by changing elements of traits such as their behavior, physiology, and morphology (termed a risk-induced trait response, or RITR); the costs to prey fitness associated with that response (termed nonconsumptive effects, or NCEs); and any resulting impacts on other species the prey interact with (termed trait-mediated indirect effects, or TMIEs). Thus, one way that warming can impact risk effects is by altering either the costs associated with responding to predators, or elements that influence the benefits of the trait response, such as the predator's foraging rates.

For example, Veldhuis et al. 2020 suggested that when prey shift their foraging to warmer but less risky times of day in response to nocturnal predators, warming may lead to stronger NCEs by making this trade-off more costly. On the other hand, if warming makes the risk vs. thermal stress trade-off too costly, prey may be forced into more frequent encounters with predators (Green et al. 2025). Other studies have suggested that warming may lead to weaker NCEs by reducing the degree to which prey overlap spatially with their predators (Barton 2010), or that warming may offset the negative effects of risk if prey are below their thermal optimum (Matassa and Trussell 2015).

The IREF is well suited for exploring how warming will impact predation risk effects, because it can explicitly capture the relationship between temperature and the costs and benefits of the prey RITR, which we can then use to determine changes in CE and NCE strength.

The general purpose of this document is to provide two generic examples outlining how we can use the IREF to explore how warming might impact predation risk effects.

Model Description

For the following scenarios, we will consider a predator-prey system where the prey can avoid predation by moving to different habitats along a temperature gradient. We will simulate this for a baseline scenario and a "warmed" scenario.

We will describe the process for deriving the mortality- and growth-trait curves below, then use those curves to predict the optimal antipredator strategy and the strength of CEs and NCEs under warming conditions.

The Mortality-Trait Curve

The mortality-trait curve is determined by the prey's likelihood of being killed, which we will define as being directly related to the predator's distribution. In this first scenario, the predator inhabits only habitats along the temperature gradient that are thermally optimal for it and does not use habitats outside that range. This is fundamentally similar to the *Daphnia-Byth* system, where *Daphnia* can migrate vertically to avoid predation at the cost of spending time in cooler, less thermally favorable environments that slow its growth.

The probability of the prey being killed is modeled using a normal curve as:

$$P_{mort}(T) = \frac{1}{\sigma_T\sqrt{2\pi}} e^{\frac{-(T-\mu_T)}{2\sigma_T^2}}$$

where T is temperature in degrees Celsius, μ_T is the mean temperature of the predator's distribution (i.e., where it peaks, the predator's thermal optimum), and σ_T is the standard deviation of that mean.

We will anchor the mean of the predator's distribution at a the far left of the plot, such that the greatest risk of predation occurs when prey are at the far left end of the temperature gradient.

```
## The mortality-trait curve sigT is the standard deviation
## of the probability distribution muT is the mean
## temperature of the probability distribution temp_C is
## the temperature at which the probability of mortality is
## being predicted
mort.curve_warming <- function(sig_T, mu_T, temp_C) {
    (1/(sqrt(2 * pi * sig_T^2))) * exp(-(((temp_C - mu_T)^2)/(2 *
        sig_T^2)))
}

sig_T.val <- 1  # Std deviation of the distribution, arbitrarily set such that p(mort) at muT is ~0.4
mu_T.val <- 20  # Mean temperature of the distribution


## Visualizing the mortality curve Define gradient of
## temperatures
temp.vec <- seq(20, 30, by = 0.1)

# Write predicted mortality to a dataframe
iref_warming_cool.df <- as.data.frame(mat.or.vec(length(temp.vec),
    2))
colnames(iref_warming_cool.df) <- c("temp_C", "mortcurve")
iref_warming_cool.df$temp_C <- temp.vec
iref_warming_cool.df$mortcurve <- mort.curve_warming(sig_T = sig_T.val,
    mu_T = mu_T.val, temp_C = temp.vec)
```

```r
# Plot it
ggplot(data = iref_warming_cool.df, aes(x = temp_C)) + geom_line(aes(y = mortcurve),
    colour = "red") + theme_classic(base_size = 10, base_line_size = 1) +
    ylab("Probability of Mortality") + xlab("Temperature (°C)") +
    scale_x_continuous(expand = c(0, 0), breaks = c(20, 25, 30)) +
    scale_y_continuous(expand = c(0, 0), limits = c(0, 1), breaks = c(0,
        0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1)
```

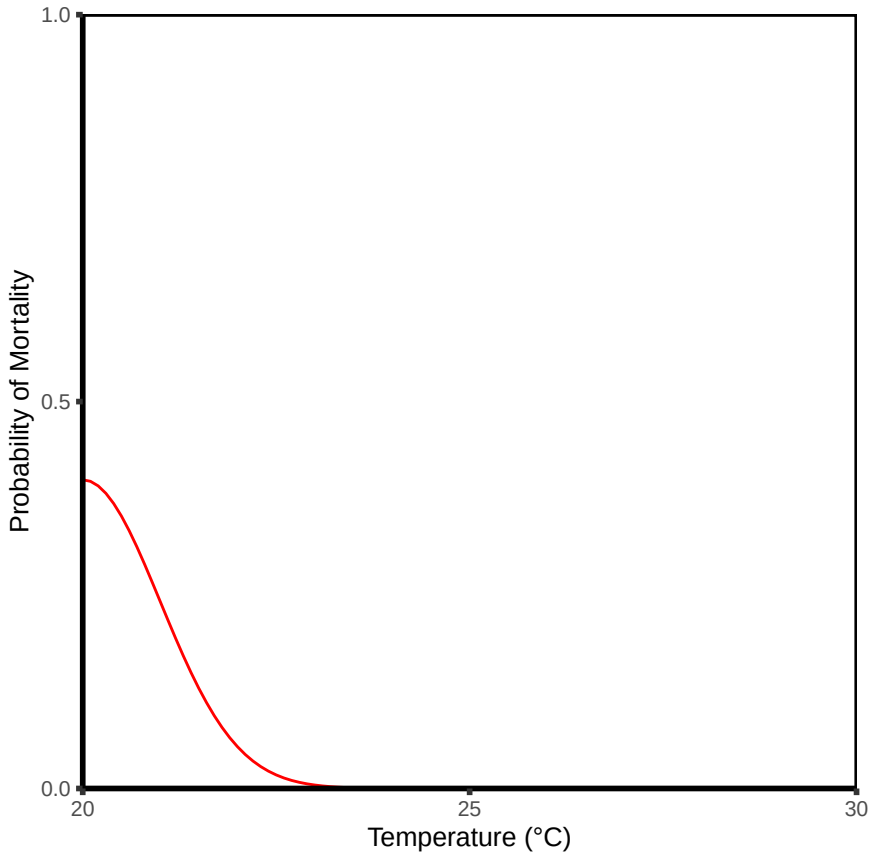

Fig. 1: The mortality-trait curve (red line), which expresses the probability of prey mortality due to predation along the temperature gradient.

The Growth-Trait Curve

Meanwhile, we will assume that the prey's growth curve follows the general form of an Arrhenius-quadratic thermal performance curve, expressed using vertex form for more convenient manipulation of the parameters:

$$f(T) = c_g e^{a_g (T - h_g)^2 + k_g}$$

where T is temperature in °C, a_g is a constant that sets the width of the curve, and the location of the curve's peak is at the point h_g, k_g. In other words, the performance curve will peak at the prey's optimal temperature h_g, and rise to a point on the graph k_g. We will set $k_g = 1$ and $c_g = 1/e$ such that prey growth is maximized at 1 when prey do not respond to predators.

```
## The growth-trait curve a.growth determines the width of
## the performance curve and should be negative, so that
## the curve is concave down h.growth determines the
## temperature at which prey growth peaks c.growth is a
## scaling constant temp_C is temperature in degrees
## Celsius
growth.curve_warming <- function(c_g, a_g, h_g, temp_C) {
    c_g * exp(a_g * ((temp_C - h_g)^2) + 1)
}

# Set a.growth to an arbitrary value that makes the curve
# broader than the predator's distribution; smaller values
# yield broader curves
a_g.val <- -0.05

# Set h.growth to the same point that the predator's
# density peaks
h_g.val <- 20

# c.growth needs to be 1/e for this parameterization to
# peak at 1
c_g.val <- 1/exp(1)

iref_warming_cool.df$growthcurve <- growth.curve_warming(c_g = c_g.val,
    a_g = a_g.val, h_g = h_g.val, temp_C = temp.vec)

iref_warming_cool.df$fitness <- 1 - (1 - iref_warming_cool.df$growthcurve) -
    iref_warming_cool.df$mortcurve
iref_warming_cool_optim.df <- iref_warming_cool.df %>%
    slice(which.max(fitness))

# Plot it
ggplot(data = iref_warming_cool.df, aes(x = temp_C)) + geom_line(aes(y = growthcurve),
    colour = "blue") + geom_line(aes(y = mortcurve), colour = "red") +
    theme_classic(base_size = 10, base_line_size = 1) + ylab("Growth and Mortality") +
    xlab("Temperature (°C)") + geom_rect(data = iref_warming_cool_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = growthcurve,
        ymax = 0.998), color = "gray2", fill = "lightblue", alpha = 0.5) +
    geom_rect(data = iref_warming_cool_optim.df, aes(xmin = temp_C -
        0.1, xmax = temp_C + 0.1, ymin = 0.001, ymax = mortcurve),
        color = "gray2", fill = "red", alpha = 0.5) + geom_rect(data = iref_warming_cool_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = mortcurve,
```

```
    ymax = growthcurve), color = "gray2", fill = "lightgreen",
alpha = 0.5) + scale_x_continuous(expand = c(0, 0), breaks = c(20,
25, 30)) + scale_y_continuous(expand = c(0, 0), limits = c(0,
1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
fill = NA, linewidth = 0.75), aspect.ratio = 1)
```

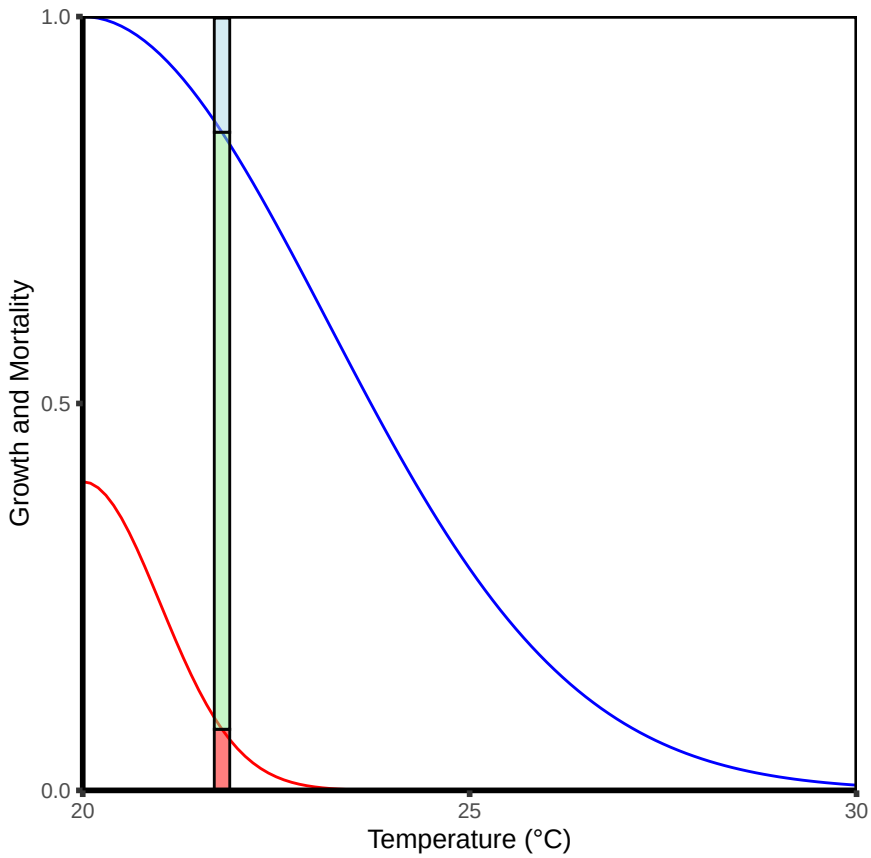

Fig. 2: Growth and mortality curves (blue and red, respectively) as a function of temperature (°C). The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength.

The optimal strategy for prey is to avoid places along the thermal gradient where predators are dense, such that the optimal habitat is around 21.8 °C. This results in a relatively weak CEs and NCEs (~0.08 and ~0.15, respectively) and high prey fitness ($1 - NCE - CE = 0.77$).

Warming Scenarios

We will now explore how warming impacts this interaction by restricting the available thermal gradient that prey can use to avoid predators. This assumes that warming impacts the entire available habitat equally, and predators remain in the same (warmed) habitat that they used before. Meanwhile, the prey's thermal performance curve is unchanged (i.e., it cannot acclimate to changes in temperature).

We will explore how this changes the interaction assuming warming increases temperatures across the gradient by ~2.5 °C.

```
# Set predator params such that mu follows a 2.5C increase
# in temperature
sig_T.val <- 1  # Std deviation of the distribution, arbitrarily set such that p(mort) at muT is ~0.5
mu_T.val <- 22.5  # Mean temperature of the distribution

temp.vec <- seq(20, 30, by = 0.1)

# Write predicted mortality to a dataframe
iref_warming_hot.df <- as.data.frame(mat.or.vec(length(temp.vec),
    2))
colnames(iref_warming_hot.df) <- c("temp_C", "mortcurve")
iref_warming_hot.df$temp_C <- temp.vec
iref_warming_hot.df$mortcurve <- mort.curve_warming(sig_T = sig_T.val,
    mu_T = mu_T.val, temp_C = temp.vec)

# Write predicted growth to a dataframe
iref_warming_hot.df$growthcurve <- growth.curve_warming(c_g = c_g.val,
    a_g = a_g.val, h_g = h_g.val, temp_C = temp.vec)

iref_warming_hot.df$fitness <- 1 - (1 - iref_warming_hot.df$growthcurve) -
    iref_warming_hot.df$mortcurve
iref_warming_hot_optim.df <- iref_warming_hot.df %>%
    filter(temp_C > 22.5) %>%
    slice(which.max(fitness))

# Plot it
ggplot(data = iref_warming_hot.df, aes(x = temp_C)) + geom_line(aes(y = growthcurve),
    colour = "blue") + geom_line(aes(y = mortcurve), colour = "red") +
    theme_classic(base_size = 10, base_line_size = 1) + ylab("Growth and Mortality") +
    xlab("Temperature (°C)") + geom_rect(data = iref_warming_hot_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = growthcurve,
        ymax = 0.998), color = "gray2", fill = "lightblue", alpha = 0.5) +
    geom_rect(data = iref_warming_hot_optim.df, aes(xmin = temp_C -
        0.1, xmax = temp_C + 0.1, ymin = 0.001, ymax = mortcurve),
        color = "gray2", fill = "red", alpha = 0.5) + geom_rect(data = iref_warming_hot_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = mortcurve,
        ymax = growthcurve), color = "gray2", fill = "lightgreen",
    alpha = 0.5) + geom_rect_pattern(xmin = 20.001, xmax = 22.5,
    ymin = 0.001, ymax = 0.998, pattern = "stripe", pattern_fill = "red",
    pattern_color = "red", fill = NA, color = "red") + scale_x_continuous(expand = c(0,
    0), breaks = c(20, 25, 30)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1)
```

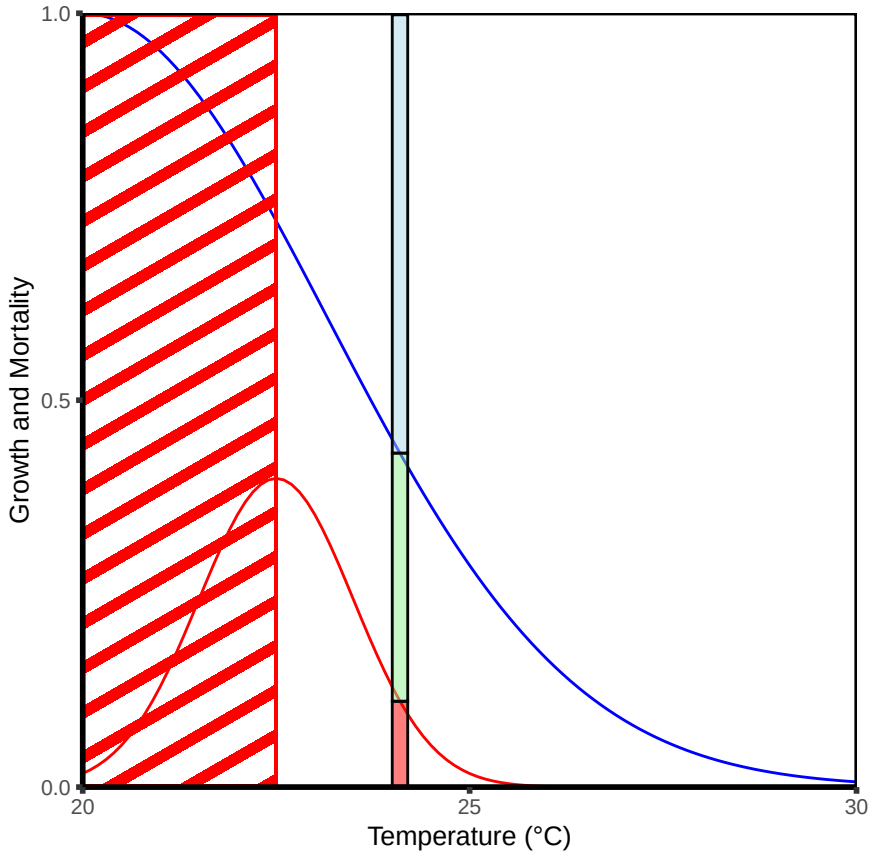

Fig. 3: Growth and mortality curves (blue and red, respectively) as a function of temperature (°C). The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength. In this scenario, the ambient temperature of the landscape has been warmed by 2.5 °C, so temperatures below the new minimum (22.5 °C) are not available for the prey to use, as indicated by the region with the red stripes.

Under these conditions, prey still respond to predators, but the optimal habitat for prey to use has shifted to the right, to ~24.1 °C. As a result, the predator's NCE is much stronger (~0.57), while the CE remains roughly the same (~0.11), and overall prey fitness is much lower ($1 - NCE - CE = 0.32$).

Biologically, this suggests that warming makes the prey's response more costly (as suggested by Veldhuis et al. 2020). Despite this increase in cost under these conditions, though, it is still worthwhile for prey to avoid habitats that are more thermally favorable to mitigate their risk of predation.

Does warming always increase NCE strength?

As established above, warming can make the prey's RITR more costly by pushing prey into habitats that are thermally unfavorable for them. Is there a point at which warming causes prey to stop responding to their predators because it is too costly, as suggested by Green et al. 2025?

Doing so using our initial parameterization of the model is tricky, because for this to occur, the slope of the growth curve will need to be steeper than the mortality curve. In other words, the loss in prey performance incurred by shifting habitats must be greater than the benefit gained by reducing predation risk.

This requires a shallower mortality curve, as I will demonstrate below by increasing the variance of the predator's distribution (σ_T^2) while keeping the maximum predation risk anchored around 0.4.

```
# Set predator params such that mu follows a 5C increase in
# temperature
sig_T.val <- 2  # Std deviation of the distribution, arbitrarily set such that p(mort) at muT is ~0.5
mu_T.val <- 22.5  # Mean temperature of the distribution

temp.vec <- seq(20, 30, by = 0.1)

# Write predicted mortality to a dataframe
iref_warming_hot2.df <- as.data.frame(mat.or.vec(length(temp.vec),
    2))
colnames(iref_warming_hot2.df) <- c("temp_C", "mortcurve")
iref_warming_hot2.df$temp_C <- temp.vec
iref_warming_hot2.df$mortcurve <- 2 * mort.curve_warming(sig_T = sig_T.val,
    mu_T = mu_T.val, temp_C = temp.vec)

# Write predicted growth to a dataframe
iref_warming_hot2.df$growthcurve <- growth.curve_warming(c_g = c_g.val,
    a_g = a_g.val, h_g = h_g.val, temp_C = temp.vec)

iref_warming_hot2.df$fitness <- 1 - (1 - iref_warming_hot2.df$growthcurve) -
    iref_warming_hot2.df$mortcurve
iref_warming_hot2_optim.df <- iref_warming_hot2.df %>%
    filter(temp_C > 22.5) %>%
    slice(which.max(fitness))

# Plot it
ggplot(data = iref_warming_hot2.df, aes(x = temp_C)) + geom_line(aes(y = growthcurve),
    colour = "blue") + geom_line(aes(y = mortcurve), colour = "red") +
    theme_classic(base_size = 10, base_line_size = 1) + ylab("Growth and Mortality") +
    xlab("Temperature (°C)") + geom_rect(data = iref_warming_hot2_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = growthcurve,
        ymax = 0.998), color = "gray2", fill = "lightblue", alpha = 0.5) +
    geom_rect(data = iref_warming_hot2_optim.df, aes(xmin = temp_C -
        0.1, xmax = temp_C + 0.1, ymin = 0.001, ymax = mortcurve),
        color = "gray2", fill = "red", alpha = 0.5) + geom_rect(data = iref_warming_hot2_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = mortcurve,
        ymax = growthcurve), color = "gray2", fill = "lightgreen",
    alpha = 0.5) + geom_rect_pattern(xmin = 20.001, xmax = 22.5,
    ymin = 0.001, ymax = 0.998, pattern = "stripe", pattern_fill = "red",
    pattern_color = "red", fill = NA, color = "red") + scale_x_continuous(expand = c(0,
    0), breaks = c(20, 25, 30)) + scale_y_continuous(expand = c(0,
    0), limits = c(0, 1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
```

```
    fill = NA, linewidth = 0.75), aspect.ratio = 1)
```

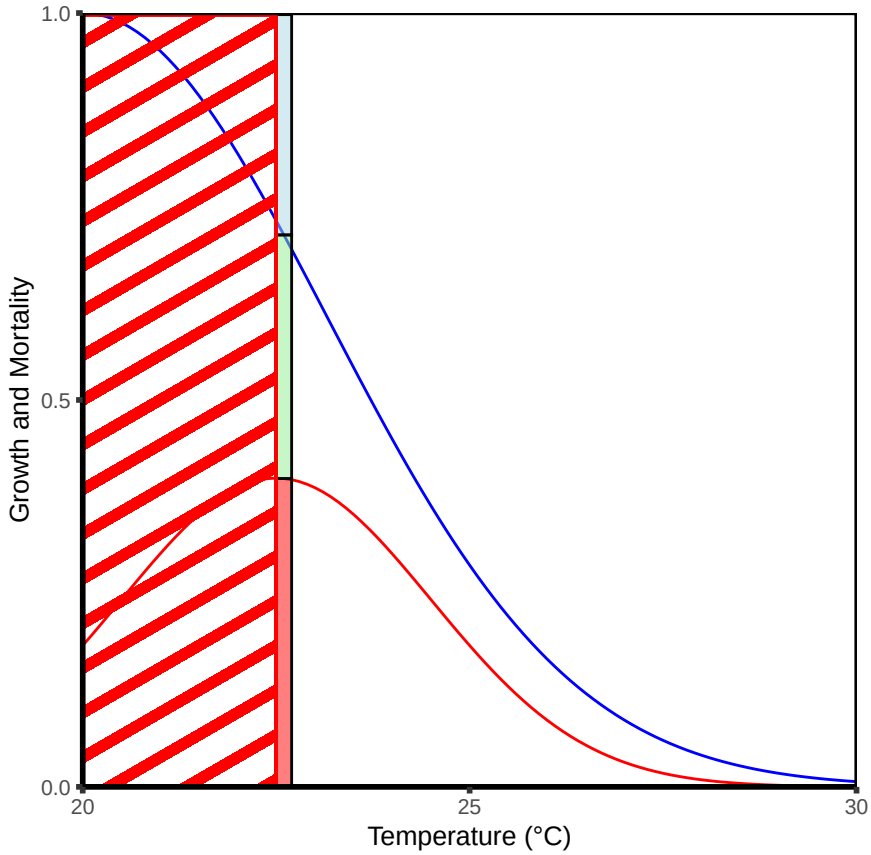

Fig. 4: Growth and mortality curves (blue and red, respectively) as a function of temperature (°C). The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength. In this scenario, the ambient temperature of the landscape has been warmed by 2.5 °C, so temperatures below the new minimum (22.5 °C) are not available for the prey to use, as indicated by the region with the red stripes.

Under these conditions, the optimal response by prey is not to avoid the predator, because it is more thermally costly than the benefit of reduced predation risk.

Biologically, this could occur if the predator is highly active across the landscape, which may be the case if the predator is relatively thermally insensitive compared to the prey (i.e., a thermal generalist, or perhaps an endotherm). Alternatively, this could occur if warming causes the predator to increase its activity across the landscape, which is a likely outcome for ectothermic predators at temperatures below their thermal optimum.

Note that NCE strength is still greater ($NCE = 0.29$) than it was in Fig. 2, but the increase in strength is more modest relative to Fig. 3.

What if warming pushes prey towards more thermally favorable conditions, instead of away from them?

The previous scenarios all operate under the assumption that warming will decrease prey growth, but this may not always be the case, especially if prey generally experience temperatures below their thermal optimum prior to warming.

If we think back to Matassa and Trussell (2015), they found a decrease in NCE strength because warming actually offset the cost of responding to predators.

Under these conditions, we will have to flip the IREF on its head and think about what will happen if warming makes the RITR *less* costly or more efficient in some way. We will explore this graphically, but let's first consider things verbally.

In these cases, I would expect that prey should generally opt to use the most thermally favorable habitat (i.e., no RITR), *unless* the increase in growth due to temperature coincided with an equal or greater increase in predation risk.

This means that for a RITR to occur, predation risk must increase with temperature (i.e., the predator distribution cannot peak below the prey's thermal optimum, because risk would then decrease as performance increases), and it must increase more rapidly than the increase in performance the prey gains.

Biologically, this could occur if the predator is highly specific in its habitat use, either due to being a thermal specialist, or due to some other environmental traits associated with habitat patches at those temperatures. A thermal generalist prey might yield similar results.

Let's explore this graphically by setting the predator and prey's thermal optima (μ_T and h_{growth}, respectively) as equal to one another, and anchoring them this time on the right side of the plot at 30°C. We will return the variance of the predator's distribution (σ_T) to 1, as in Figs. 1-2.

```
# Set predator params such that mu follows a 5C increase in
# temperature
sig_T.val <- 1  # Std deviation of the distribution, arbitrarily set such that p(mort) at muT is ~0.5
mu_T.val <- 30  # Mean temperature of the distribution

temp.vec <- seq(20, 30, by = 0.1)

# Write predicted mortality to a dataframe
iref_warming_inverted.df <- as.data.frame(mat.or.vec(length(temp.vec),
    2))
colnames(iref_warming_inverted.df) <- c("temp_C", "mortcurve")
iref_warming_inverted.df$temp_C <- temp.vec
iref_warming_inverted.df$mortcurve <- mort.curve_warming(sig_T = sig_T.val,
    mu_T = mu_T.val, temp_C = temp.vec)


# Set a.growth to an arbitrary value that makes the curve
# broader than the predator's distribution; smaller values
# yield broader curves
a_g.val <- -0.05

# Set h.growth to the same point that the predator's
# density peaks
h_g.val <- 30

# c.growth needs to be 1/e for this parameterization to
# peak at 1
```

```r
c_g.val <- 1/exp(1)

# Write predicted growth to a dataframe
iref_warming_inverted.df$growthcurve <- growth.curve_warming(c_g = c_g.val,
    a_g = a_g.val, h_g = h_g.val, temp_C = temp.vec)

iref_warming_inverted.df$fitness <- 1 - (1 - iref_warming_inverted.df$growthcurve) -
    iref_warming_inverted.df$mortcurve
iref_warming_inverted_optim.df <- iref_warming_inverted.df %>%
    slice(which.max(fitness))

# Plot it
ggplot(data = iref_warming_inverted.df, aes(x = temp_C)) + geom_line(aes(y = growthcurve),
    colour = "blue") + geom_line(aes(y = mortcurve), colour = "red") +
    theme_classic(base_size = 10, base_line_size = 1) + ylab("Growth and Mortality") +
    xlab("Temperature (°C)") + geom_rect(data = iref_warming_inverted_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = growthcurve,
        ymax = 0.998), color = "gray2", fill = "lightblue", alpha = 0.5) +
    geom_rect(data = iref_warming_inverted_optim.df, aes(xmin = temp_C -
        0.1, xmax = temp_C + 0.1, ymin = 0.001, ymax = mortcurve),
        color = "gray2", fill = "red", alpha = 0.5) + geom_rect(data = iref_warming_inverted_optim.df,
    aes(xmin = temp_C - 0.1, xmax = temp_C + 0.1, ymin = mortcurve,
        ymax = growthcurve), color = "gray2", fill = "lightgreen",
    alpha = 0.5) + scale_x_continuous(expand = c(0, 0), breaks = c(20,
    25, 30)) + scale_y_continuous(expand = c(0, 0), limits = c(0,
    1), breaks = c(0, 0.5, 1)) + theme(panel.border = element_rect(color = "black",
    fill = NA, linewidth = 0.75), aspect.ratio = 1)
```

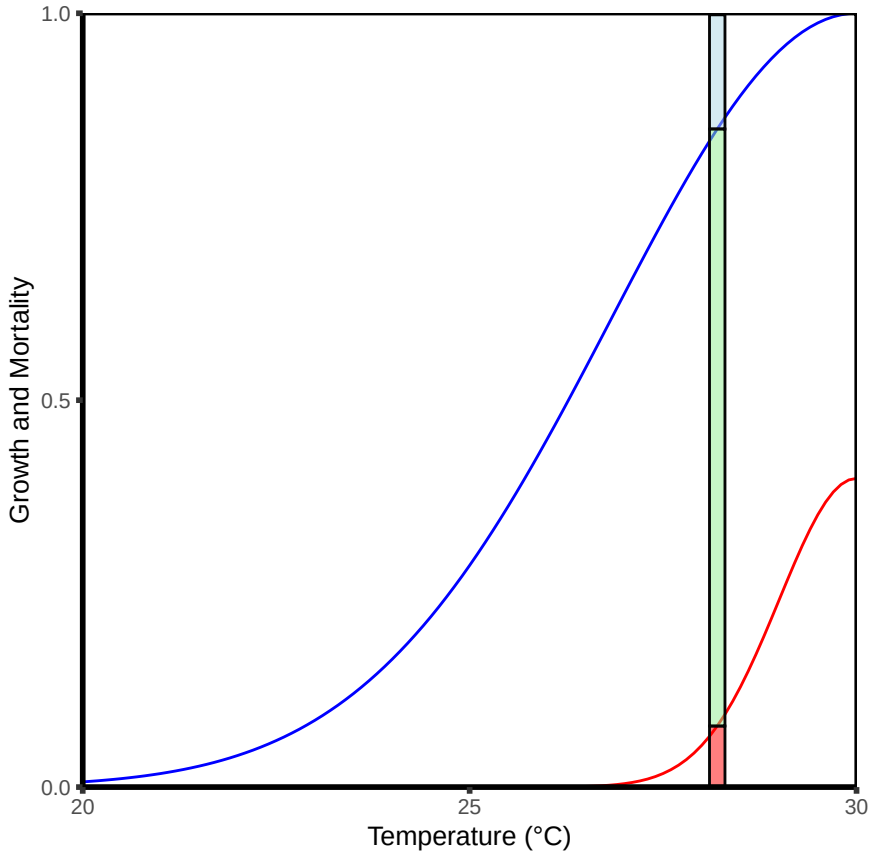

Fig. 5: Growth and mortality curves (blue and red, respectively) as a function temperature (°C). The optimal RITR is noted using a bar, where the green shaded region is fitness (1 - NCE - CE), the blue shaded region is NCE strength, and the red shaded region is CE strength.

Given the above conditions, prey should not use the most thermally favorable environments at 30 °C, because the increase in risk outweighs the benefits of better thermal performance. Further, warming would exacerbate this problem by restricting prey to habitats further to the right side of the plot, where CEs would continue to increase at a faster rate than NCEs (i.e., fitness will always be lower).